Rishika Bera

Evanston, IL | berarishika@gmail.com | linkedin.com/in/rishika-bera-05b7b025a
Portfolio: rishika2024.github.io | Citizenship: United States

EDUCATION

Northwestern University — Evanston, IL
MS in Robotics — Expected Aug 2026 - Dec 2026

Indian Institute of Technology - Jodhpur — Rajasthan, India
B.Tech in Mechanical Engineering — Dec 2021 – May 2025

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB, HTML, CNC G-Code
Robotics: ROS 2, UAV, SLAM, Path Planning, Control Systems, Perception, Sensor Fusion, Motion Planning
Machine Learning & AI: Machine Learning, MLP, CNNs, Autoencoders, Deep Learning, LLMs
Software & Libraries PX4, MoveIt, OpenCV, Gazebo, RViz, CoppeliaSim, Unity3D, Simulink
Tools: Linux, Git
Hardware: Pixhawk, Franka Arm, Interbotix Arm, RealSense Camera, Arduino, Raspberry Pi
Design & Manufacturing: SolidWorks, Onshape, Blender, 3D Printing, CNC Machine, Soldering

PROJECTS

Morphing Aerial–Ground Transformer Robot | (CAD, ROS 2, PX4, Raspberry Pi, UAVs) — Jan 2026 - Present
- Designing and CAD-modeling a transformer robot capable of both aerial flight and ground locomotion
- Assembled and tested a quadrotor prototype under manual/teleoperated control.
- Developing a mode-switching system using Raspberry Pi, ROS 2, and PX4 to transition between aerial and ground operation

Bug-Sorting - Franka Robot Arm | (Python, ROS 2, OpenCV) — Dec 2025
- Contributed to developing an autonomous system to catch and sort moving HexBugs with the Franka Arm
- Built a vision system using OpenCV and RealSense to detect, label, and track HexBugs in real-time
- Explored continuous tracking and pick-and-place motion planning using ROS 2 and MoveIt
- Collected trajectory data for ML-based motion prediction integration

Simulating Dynamics of a Jack inside a moving Box | (Dynamics, Python) — Dec 2025
- Developed a Python simulation of a jack in a moving box by deriving Euler-Lagrange equations
- Implemented collision handling with time-varying forces to model realistic motion

Grab the pen - PincherX Robot Arm | (Python, OpenCV, ROS 2) — Dec 2025
- Developed a vision and manipulation pipeline to identify objects and pick it up using Python and Realsense
- Used OpenCV to apply a color-based filter (HSV, depth) on live image and identify the contours
- Calibrated camera-robot transform for pick and place

Underwater Manipulator | (SolidWorks, 3D Printing, MATLAB, Simulink, Arduino) — Oct 2024 - Dec 2024
- Designed a rack and pinion gripper mechanism in SolidWorks and fabricated it using 3D printing
- Developed pick-and-place algorithms in MATLAB and deployed to Arduino via Simulink
- Integrated encoders for position feedback and implemented PID tuning for precise actuation

WORK EXPERIENCE

Purdue University — May 2024 - Jul 2024
Summer Undergraduate Research (SURF) Intern
- Engineered PrimerCurator's multimodal backend pipeline integrating text, image, and video using ChatGPT API
- Reduced manual tutorial creation time by 78% with >90% accuracy in automated extraction
- Collaborated with research team to ensure scalable performance and presented system design and results